Production For Firmware Download Guide

ID: RK-SM-YF-180

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DISCLAIMER

Trademark Statement

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Preface

Overview

This document introduces production solution of firmware download for the RK platform, including how to create a burning image, the use of burning tools, and common problem handling.

Product Version

Chipset	Kernel Version
RK3326	Linux4.4, Linux4.19
RK3399	Linux4.4, Linux4.19
RK3368	Linux4.4, Linux4.19
RK3288	Linux4.4, Linux4.19
RK3328	Linux4.4, Linux4.19, Linux3.10

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

** Number**	Author	Date	Change Description
V1.0.0	Liu Yi	2016-07-18	Initial draft
V1.1.0	Liu Yi	2017-02-14	Added support for RK3328
V1.2.0	Liu Yi	2019-11-13	Added support for Linux4.19
V1.2.1	Huang Ying	2021-05-19	Modified format
V1.2.2	Liu Yi	2022-05-26	ID From RK-SM-YF-179 to RK-SM-YF-180
V1.2.3	Zhao Yifeng	2023-02-09	Modified EXT CSD configuration description

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Step 1: Create the update.img firmware for upgrade

Step 2: Use FactoryTool for batch flashing

1.2 SD Upgrade Plan

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Step 2: Use the SD_Firmware_Tool to create an SD card for firmware upgrade

Step 3: Insert the upgrade SD card, power up again, and perform firmware burning

1.3 Firmware Upgrade Plan (Burner Upgrade Plan)

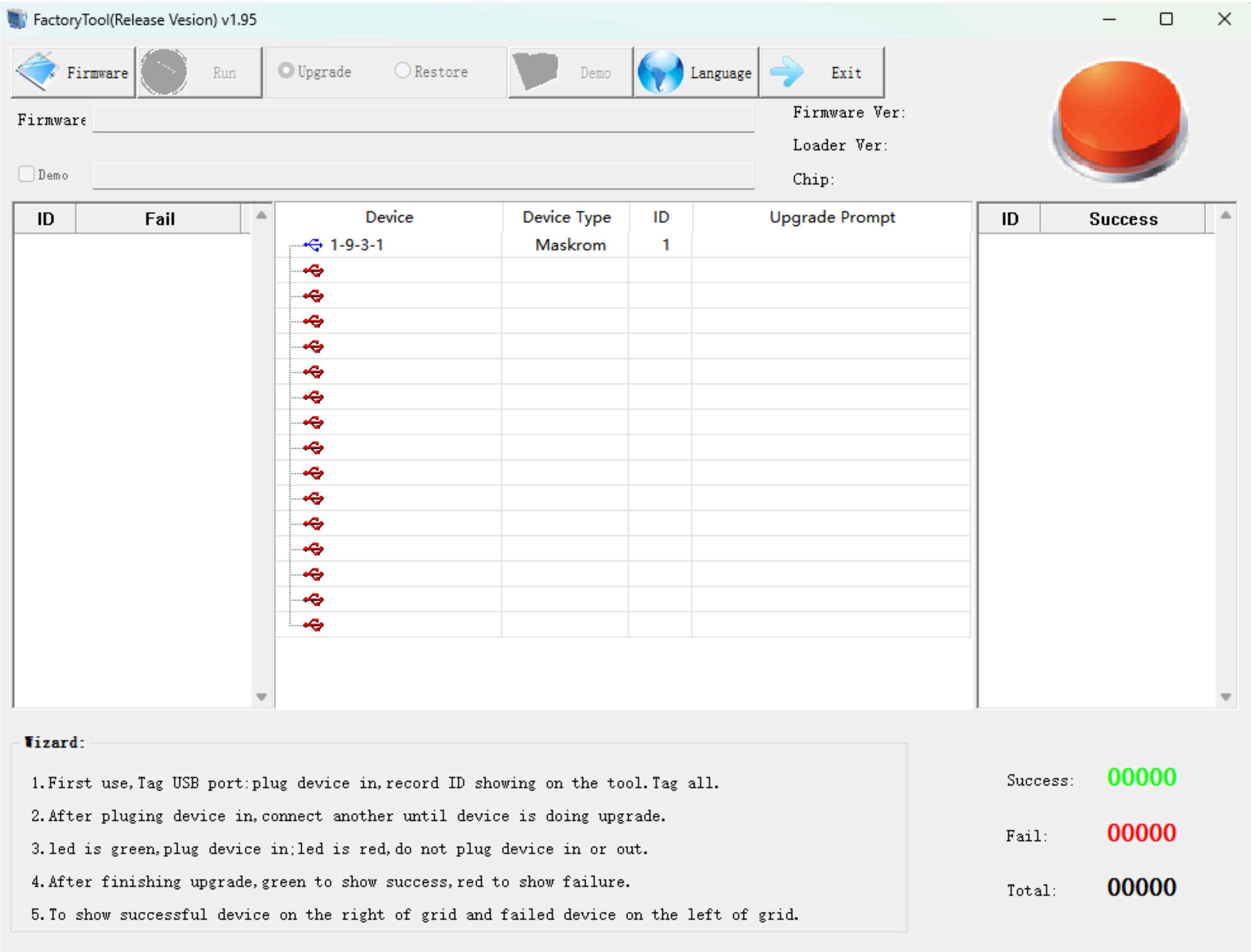
Step 1: Create the update.img firmware upgrade package

Step 2: Use the SpiImageTool tool to create the burner's firmware burning file

Step 3: Connect the storage chip to the burner and perform firmware burning

2. Tool Usage

2.1 FactoryTool Batch Flashing Tool



Usage Steps:

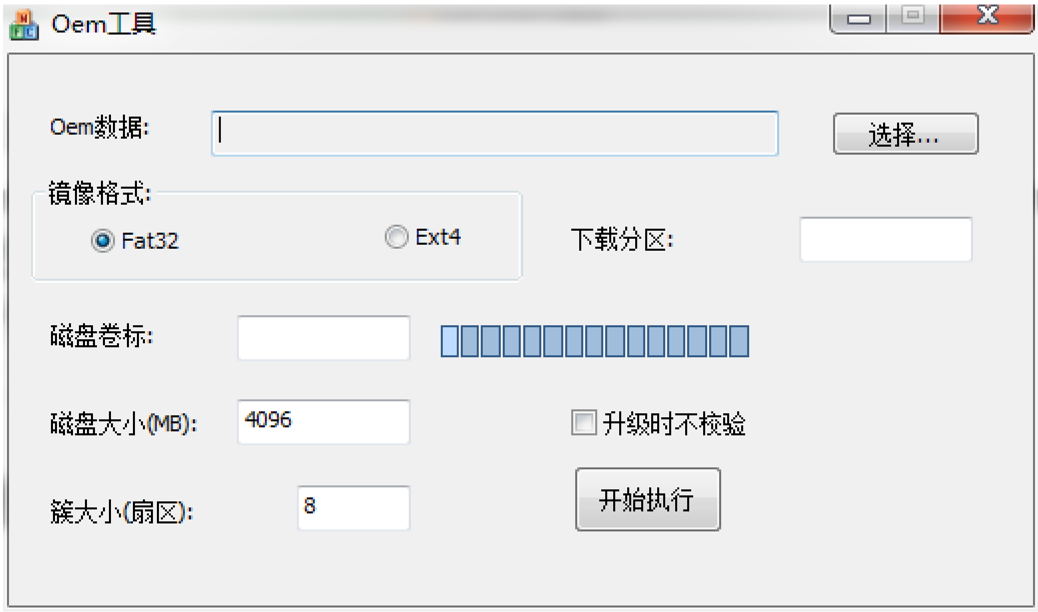
Click on "Firmware" to select the upgrade firmware.

If there is a burning Demo image, check the "Demo" option to select the Demo image (optional). For the creation of Demo images, refer to the OemTool tool usage.

Click "Start" to begin the automatic detection and upgrade of the device.

Connect the device for upgrade, once the tool detects it, the upgrade will start automatically.

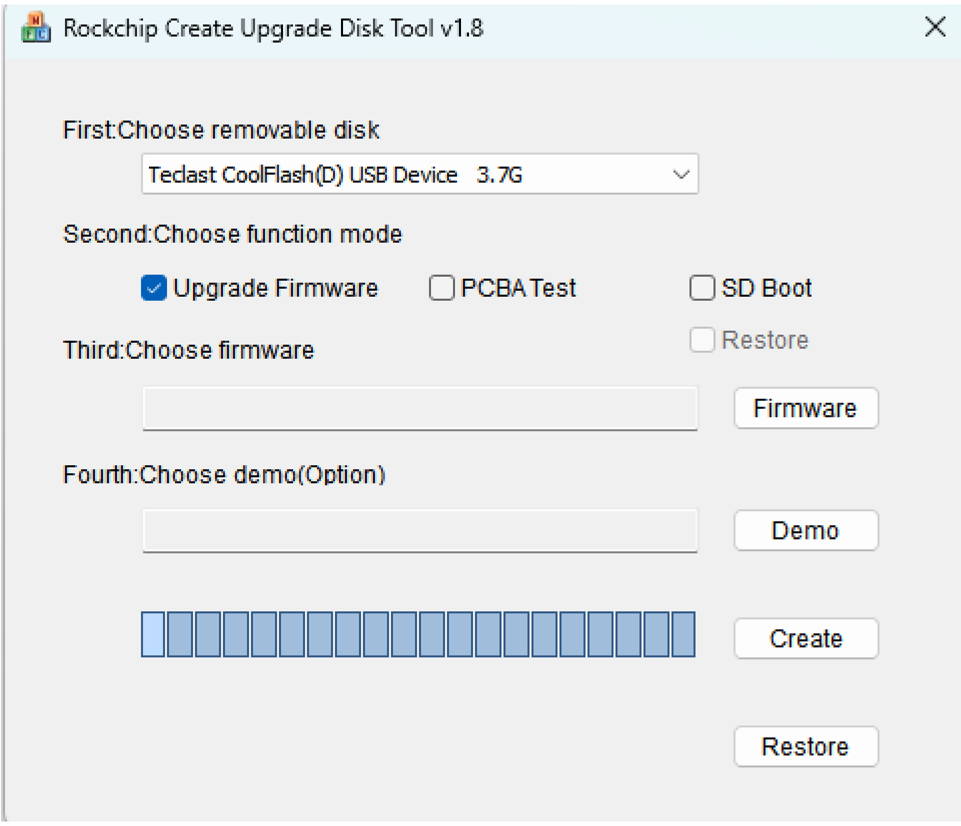
2.2 OemTool (Demo Image Creation Tool)



Steps for Creating a Demo Image:

1. Click "Select..." to choose the Demo directory for image creation.
2. Check "Fat32", as only Fat32 format images are currently supported.
3. Set "Disk Size" to be greater than the user partition capacity, aligned in 100M increments.
4. Click "Start Execution", and upon success, an OemImage.img image file will be generated in the tool directory.

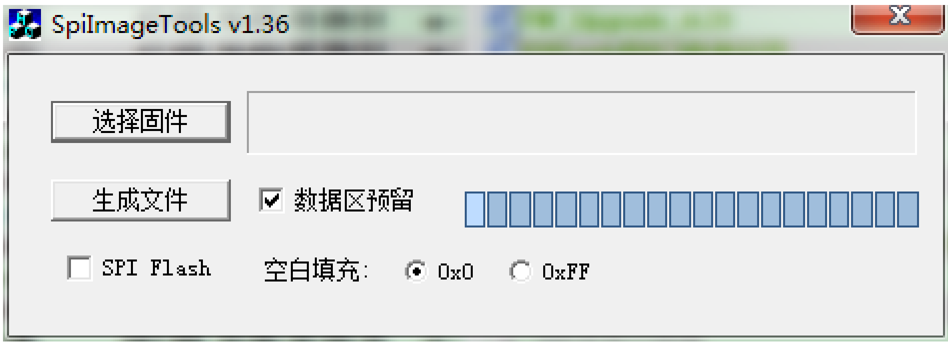
2.3 SD Firmware Tool (SD Upgrade Card Creation Tool)



Steps for Creating an SD Upgrade Card:

1. Select the SD card or USB drive to be created from the drop-down list.
2. Check the box for "Firmware Upgrade".
3. Click "Select Firmware" to choose the update.img upgrade firmware.
4. Click "Start Creation".

2.4 SpiImageTool (Firmware Image Creation Tool)



Firmware Image Creation Steps:

1. Click "Select Firmware" to choose the update.img upgrade firmware.
2. When using Emmc for storage, check the "Data Area Reservation" option.
3. When using Emmc for storage, select 0 for blank filling; when using nandflash, select 0xFF for blank filling.

4. Click "Generate File". Upon success, boot0.bin and data.bin will be generated in the tool directory. For Emmc, only data.bin is used; for nandflash, both boot0.bin and data.bin are required.

3. Creating Firmware Upgrades

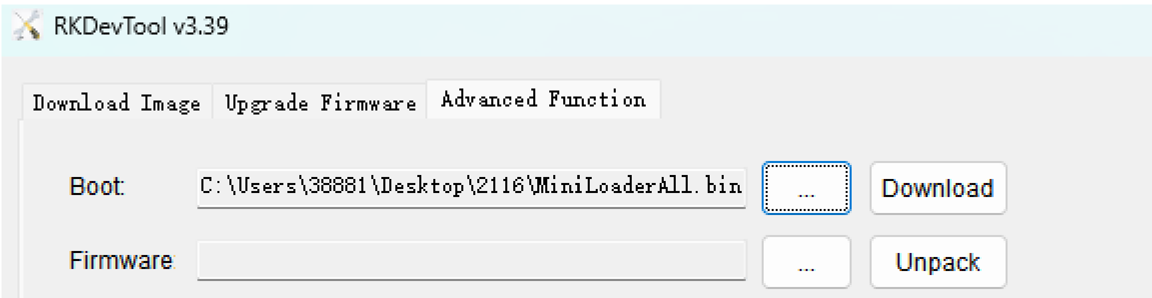
3.1 Steps

1. Navigate to the Android source code directory and execute the `mkimage.sh` script with the `ota` parameter to generate `system.img`, `boot.img`, and `recovery.img`, among others. Copy these images to the `image` directory under `rockdev`.
2. In the `rockdev` directory of AndroidTool, run the `mkupdate.bat` batch file to generate the `update.img` upgrade firmware. On Ubuntu, execute the `mkupdate.sh` script to generate the same. The following image shows the content of `mkupdate.bat`:

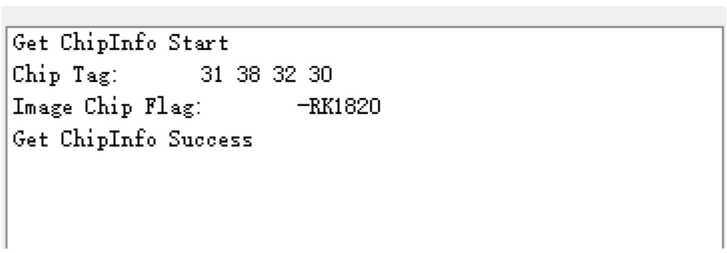
```
Afptool -pack ./ Image\update.img


RKImageMaker.exe -RK31 RK3188Loader(L)_V2.10.bin  Image\update.img update.img -os_type:androidos

```

Pay special attention to the `-RK31` parameter, which needs to match the device. If you are unsure about this value, you can obtain it through the following methods:

- Launch the `androidtool` utility, access advanced features, select the loader file for this scheme, and click on "Download".



- Click on "Read Chip Info" at the bottom, and the following information will be printed on the right side, with `Image Chip Flag` being the parameter you need.



4. Burner Settings

4.1 EMMC Flashing Data

EMMC is divided into three parts: the USER area, BOOT1 area, and BOOT2. It is only necessary to flash the USER partition, and the file to be flashed is the data.bin generated by SpiImageTool.

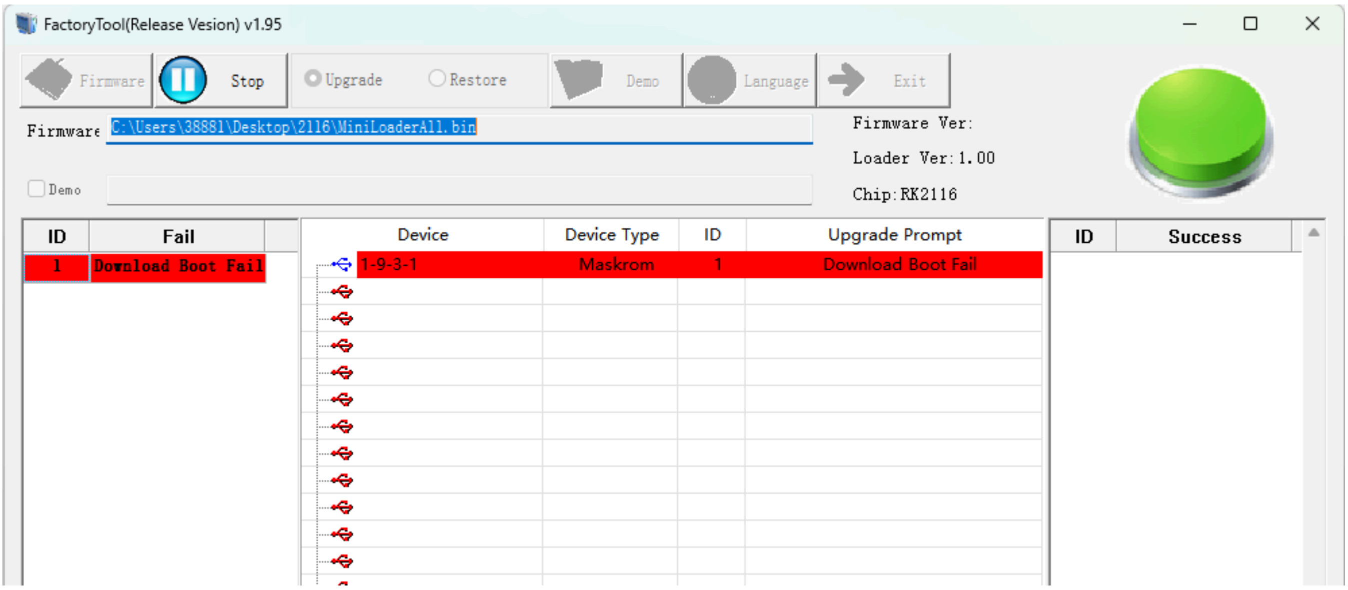
4.2 EMMC EXT CSD Configuration Information

All values are set to default and do not require configuration.

```
EXT_CSD[167] = 0x1f (default value)

EXT_CSD[162] = 0x0 (default value)

EXT_CSD[177] = 0x0 (default value)

EXT_CSD[178] = 0x0 (default value)

EXT_CSD[179] = 0x0 (default value)
```

5. Common Upgrade Issues

5.1 Downloading Boot Failure I



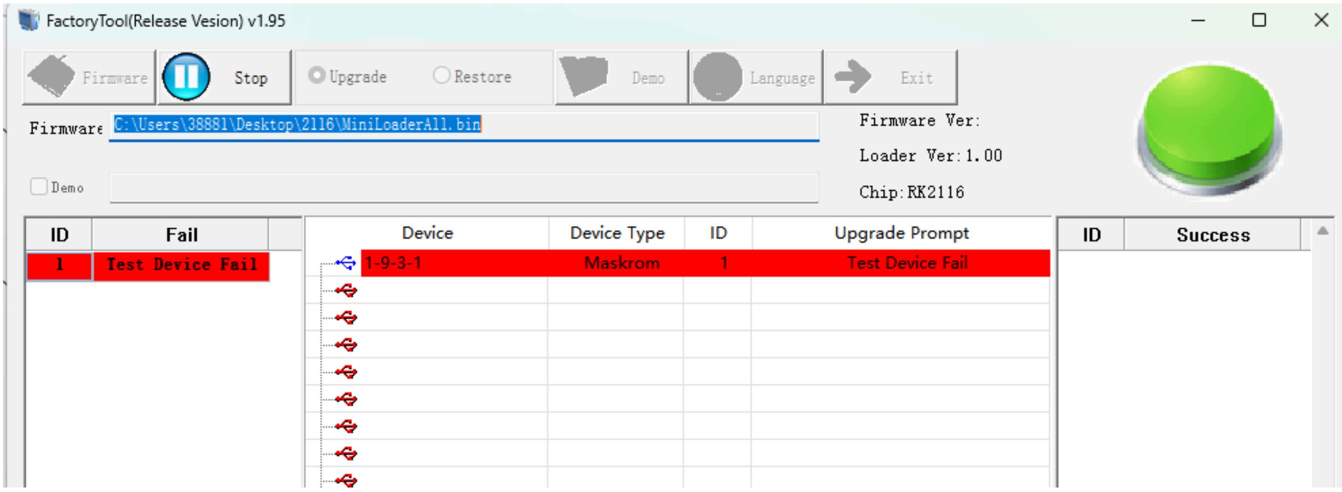
Log as below:

```
14:41:15 275    Layer<1,1-9-3-1>:Download Boot Start
14:41:20 334    <LAYER 1-9-3-1> ERROR:Boot_VendorRequest-->DeviceIoControl failed,Total(2050),Sended(0),crc(959b),err(121)
14:41:20 334    <LAYER 1-9-3-1> ERROR:DownloadBoot-->Boot_VendorRequest471 failed,index(0)
14:41:20 344    [Error] Layer<1,1-9-3-1>:Download Boot Fail
```

Possible Causes:

1. Poor USB Signal (Check if the capacitors and resistors on the USB line are normal, and if the USB power supply is normal)
2. Controller Cold Solder or Power Supply Issue

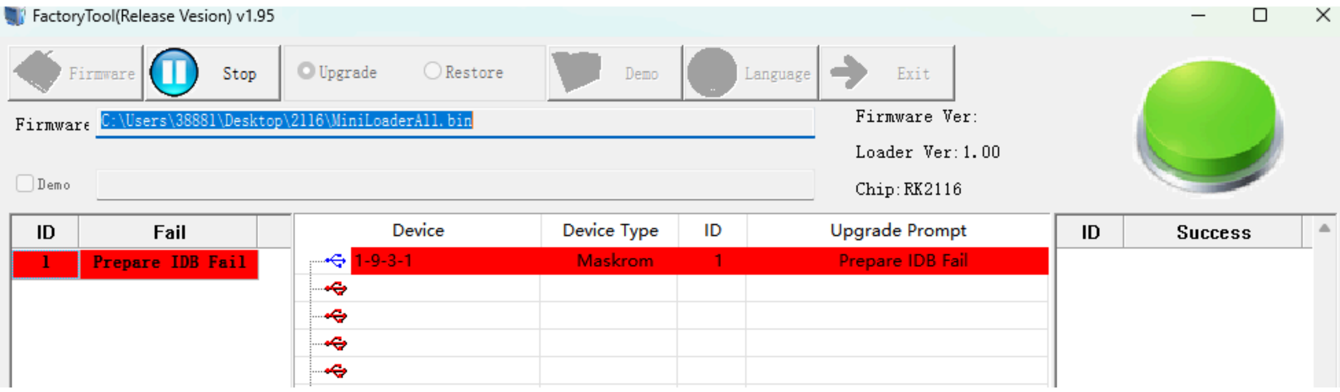
5.2 Downloading Boot Failure II



Possible Causes:

DDR chip or routing issue

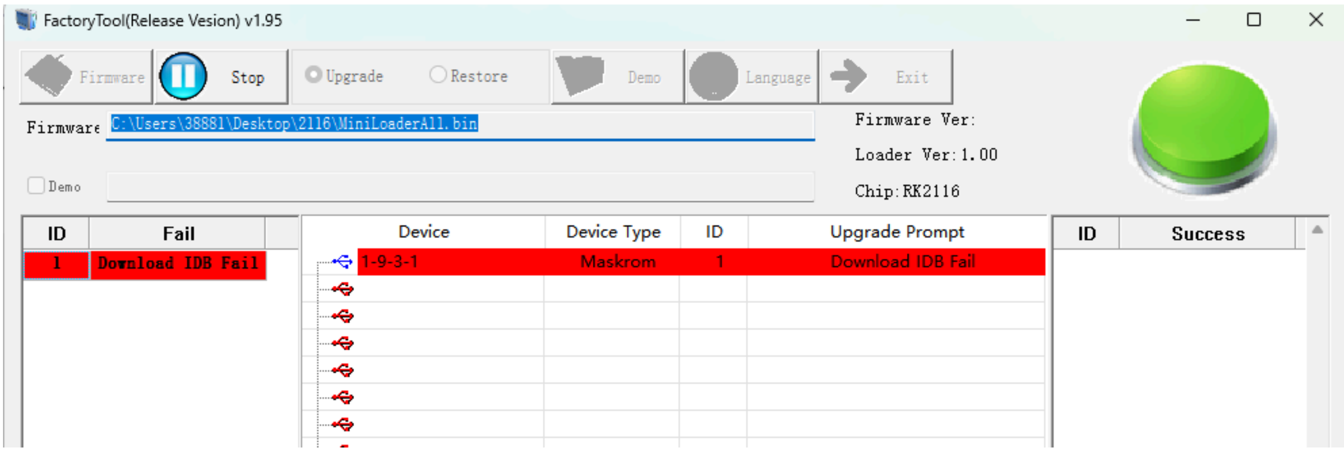
5.3 Preparing IDB Failure



Possible Causes:

Soldering issues on the Flash or unsupported chip types

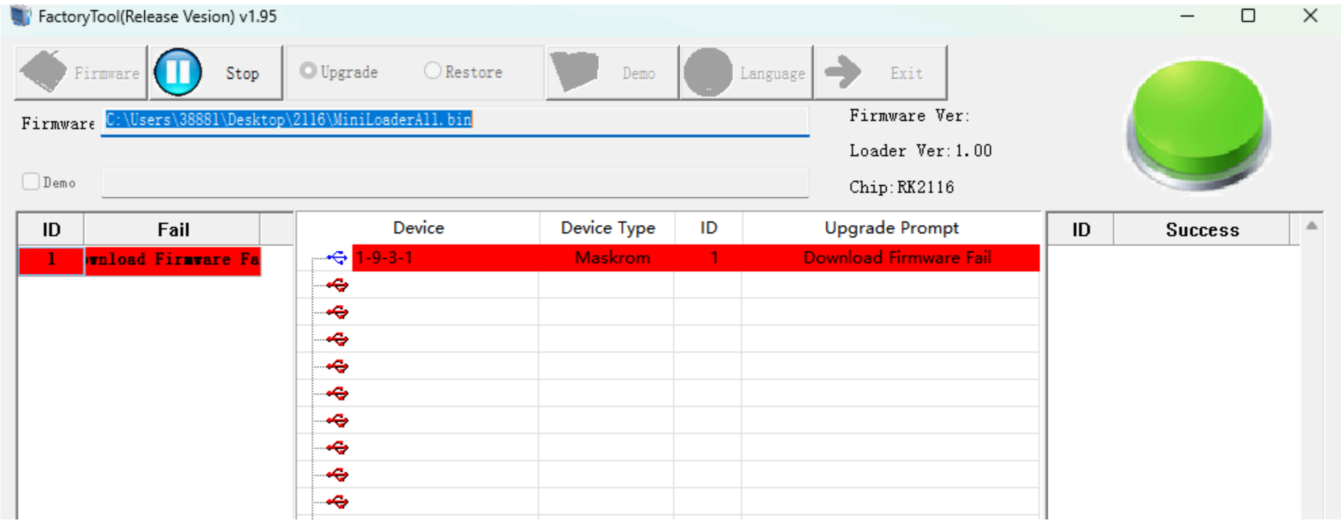
5.4 Failed to Download IDB



Possible Causes:

1. USB Communication Issues (Retry after power cycling, use a powered USB hub)
2. DDR Stability Issues (Perform stability tests using DDR testing tools)

5.5 Firmware Download Failure



Possible Causes:

1. USB Communication Issues (Retry after power cycling, use a powered USB hub)
2. Flash Issues (Erase flash using AndroidTool and retry)